Malhar Gudekar

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Education

University of Illinois Urbana Champaign — August 2024 - May 2026
Master of Science, Information Management — GPA: 3.8/4
• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai — June 2019 - May 2023
Bachelor of Engineering, Computer Science — GPA: 9.25/10
• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

Skills

• **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation
• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM, R, Scipy, Statsmodels, Scikit-learn
• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP, Jenkins, Apache Spark

Work Experience

University of Illinois Urbana Champaign — May 2025 - Present
Research Assistant - iSchool
- Designed and implemented end-to-end machine learning pipeline to analyze behavioral data from 100+ users, applying feature engineering, segmentation techniques, and statistical transformations to generate actionable insights for downstream product and operations teams
- Designed relational data models and database schema for high-volume datasets, applying segmentation and indexing strategies that improved query performance by 38% and enabled faster feature extraction for machine learning workflows
- Built embedding-based retrieval and summarization pipeline to enhance contextual understanding of user interactions, improving multi-turn response accuracy by 43% and supporting more reliable behavioral analysis

Research Assistant - CHI AI in Infodemic Management — January 2025 - May 2025
- Developed NLP-based machine learning models in Python to classify and analyze large-scale text datasets, applying statistical preprocessing, feature extraction, and model optimization to reduce processing latency by 41% and improve classification accuracy
- Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets
- Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group — August 2025 - December 2025
Technical Consultant
- Built AI-driven document analysis system using Python, embeddings, and statistical retrieval techniques, combining feature engineering and optimization to improve information extraction efficiency by 64%
- Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases
- Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider — July 2023 - October 2023
Data Analyst
- Analyzed large-scale operational datasets using Python, PySpark, and Spark distributed processing to identify trends, anomalies, and performance issues, applying statistical analysis to surface actionable insights for operations and business stakeholders
- Visualized statistical findings and KPIs in Power BI and Tableau dashboards, translating model outputs into clear, actionable reports for operations and executive stakeholders, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd. — January 2023 - June 2023
Data Analyst
- Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.
- Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois — August 2025 - Present
Project Manager
- Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

Projects

Scalable ML System for Customer Risk Prediction
- Built **LightGBM**-based machine learning model to **predict customer credit risk** from **time-series transaction data**, combining feature engineering, classification modeling, and evaluation techniques to identify high-risk and low-risk segments.
- Performed statistical feature engineering on behavioral patterns including transaction frequency and spending trends, using cross-validation and statistical evaluation to improve model precision by 20% and support interpretable, results-oriented recommendations
- Derived actionable insights on pricing, promotions, and stock behavior using R and Python, enabling inventory planning decisions that reduced stockout and overstock risks across five categories